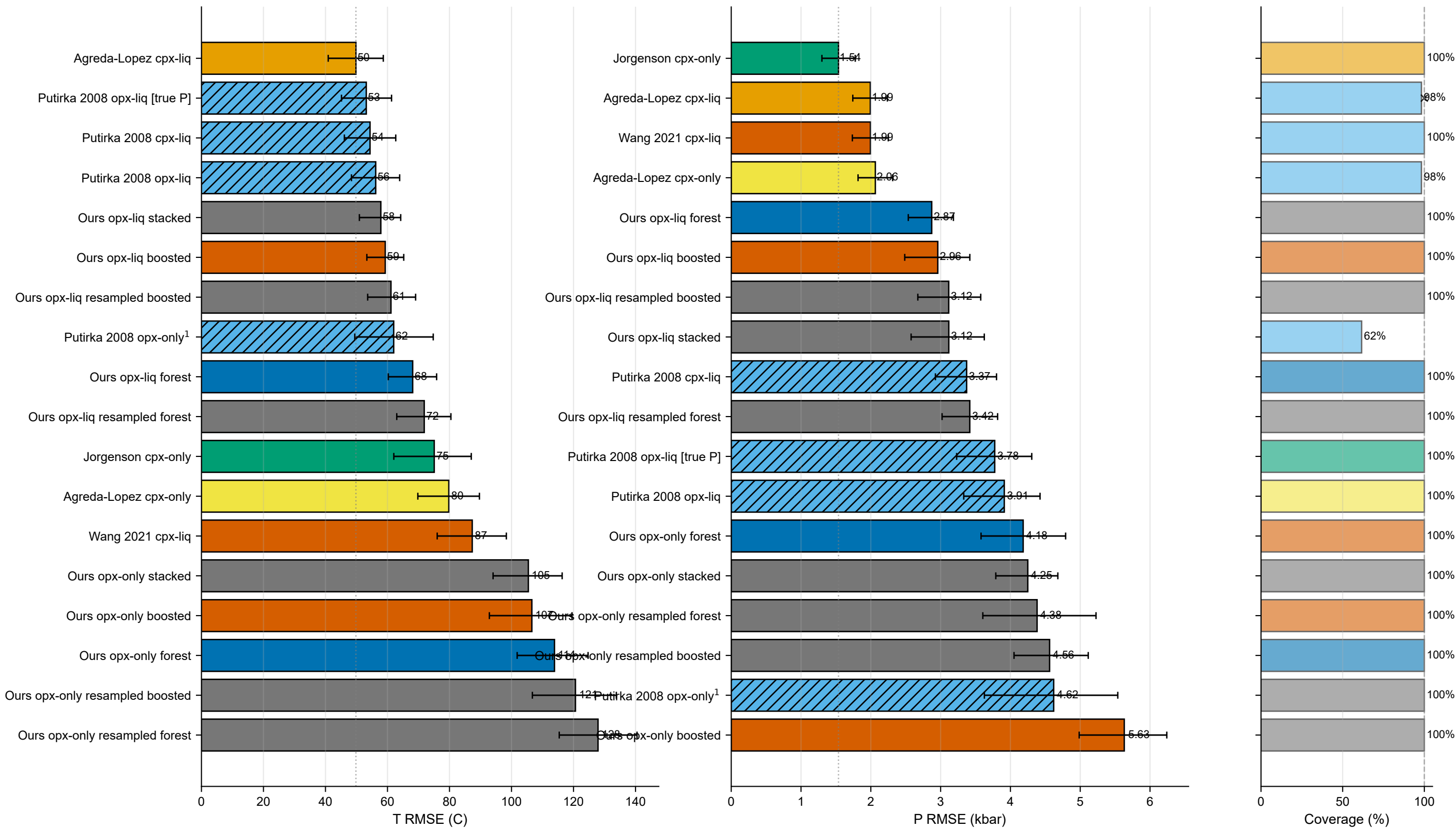


Method benchmark — ArcPL three-phase paired (cpx + opx + liq) (n_total = 112)



Bar height (panels A, B): RMSE with 95% bootstrap CI (B=2000). Bar height (panel C): coverage = min(T, P) finite-prediction rate; x marker = higher of T/P if differs. Hatched bars = classical (Putirka 2008); solid bars = ML methods. Methods sorted by T RMSE ascending.

¹ Putirka 2008 opx-only: 62% coverage (limited by P missing-row filter)